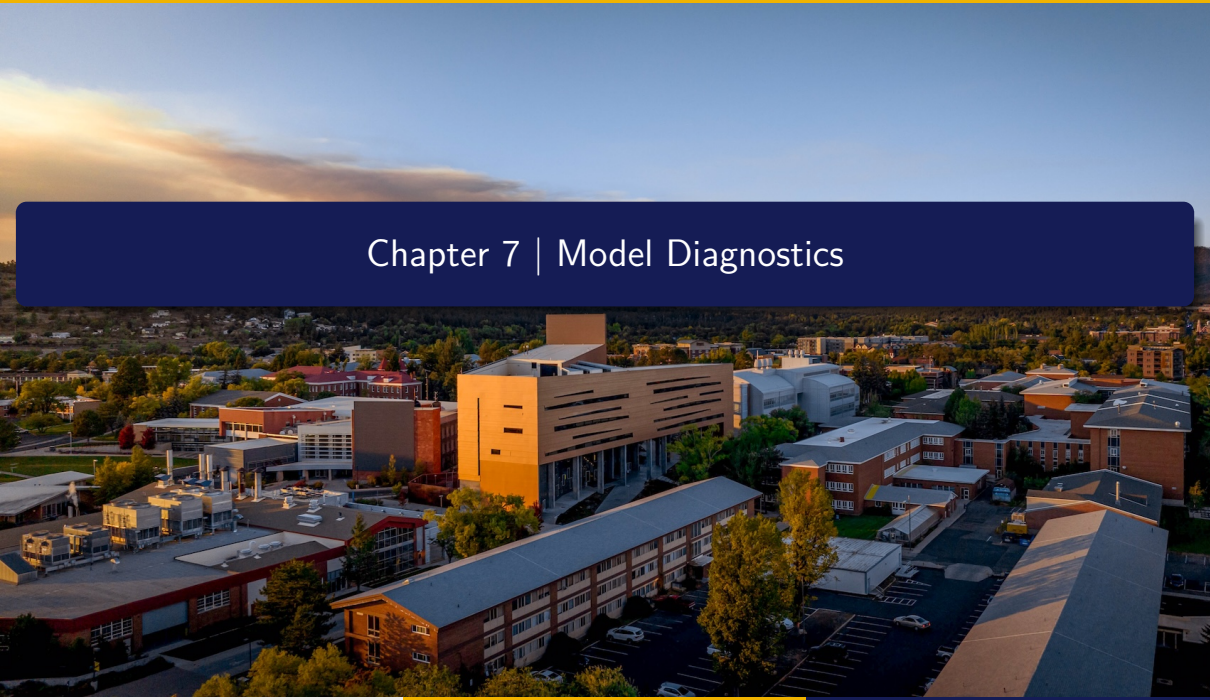
Chapter 7 | Model Diagnostics



Learning objectives

- We need to assess whether our linear model is a good model and whether the data violate any required assumptions.
- Upon successful completion of this chapter, we will be able to do the following tasks:
  1. Influential points identification: Unusual observations could contribute too much influence to the analysis. These few influential observations might drastically change the outcome of the model.
  2. Model specification assessment: Our model specification $E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$ might be wrong, and we might need to include different predictors to the model.
  3. Model error distribution verification: We have assumed that $\epsilon_i \overset{\text{iid}}{\sim} \text{N}(0, \sigma^2)$, but autocorrelation, heteroscedasticity, and non-normality might be present.
- Note: Problems with one of these can be often corrected by transforming either the explanatory or response variables.

Outline

1. Introduction
2. Model Specification
3. Influential Observations
4. Model Error Assumptions

Three potential issues

- Once we fitted a linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, there are three potential issues.
- Issue 1: Model misspecification. Our model specification $\mathrm{E}(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$ might be wrong, and we might want to include other predictors (or remove existing predictors from) in the model to improve model fitness.

Three potential issues

- Issue 2: Influential points. Some observations might influence too much to the fitted model. These observations could drastically change the outcome of the model.

Three potential issues

- Issue 3: Violation on the model error assumptions. We assumed $\boldsymbol{\varepsilon} \sim \mathrm{N}(\mathbf{0}, \sigma^2\mathbf{I})$. Specifically, $\epsilon_i \overset{\text{iid}}{\sim} \mathrm{N}(0, \sigma^2)$ for SLR or $\epsilon_{ij} \overset{\text{iid}}{\sim} \mathrm{N}(0, \sigma^2)$ for ANOVA.
  1. Independent? The responses y_i and y_j could be correlated (especially for time series).
  2. Identically distributed? The errors ϵ_i's might show heteroscedasticity (or an unequal variance).
  3. Normally distributed? Nonnormality might be present. (A remedy could be a transformation of the response variable.)

Specification of mean response via data plots

- Correct specification of the mean response $\mathrm{E}(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$ is crucial.
- How do we know if a model is good or not? A simple method is to plot the data.
  1. Plots of the response $\mathbf{y}$ vs. k-th predictor $\mathbf{x}_k$ help us to build a correct model for $\mathrm{E}(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$.
  2. These plots also help us to identify a categorical variable to include.
  3. Often ignored is measurement error. If the predictor is perturbed with measurement errors, the least squares estimates for $\boldsymbol{\beta}$ could be incorrect.

Specification of mean response via residuals

- Once a model was fitted, we can also assess the fitted model based on residuals:

$$r_i = \hat{\epsilon}_i = y_i - \hat{y}_i \quad \text{for } i = 1, \ldots, n$$

- If the fitted model is good, then
  1. Plot of $\{(\hat{y}_1, r_1), \ldots, (\hat{y}_n, r_n)\}$ does not show any pattern.
  2. Plot of $\{(x_{1k}, r_1), \ldots, (x_{nk}, r_n)\}$ does not show any pattern for each predictor $\mathbf{x}_k$.
  3. Plot of $\{(y_1, \hat{y}_1), \ldots, (y_n, \hat{y}_n)\}$ approximately lies on a linear line with intercept 0 and slope 1.

Methods of identifying influential observations

- How do we identify influential observations?
  1. Plot the data $\{(x_{1k}, r_1), \ldots, (x_{nk}, r_n)\}$ for each predictor $\mathbf{x}_k$.
  2. Plot $\{(\hat{y}_1, r_1), \ldots, (\hat{y}_n, r_n)\}$.
  3. Compute the standardized residuals.
  4. Compute Cook's Distance.

Residuals | Review

- In SLR $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, the fitted value (predicted response) $\hat{y}_i$ is $\boxed{\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i}$ for $i = 1, \ldots, n$, or

$$\begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} \Rightarrow \boxed{\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}}$$

  1. Because $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y}$, we express $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}[(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y}] = \mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y} = \mathbf{H}\mathbf{y}$.
  2. $\mathbf{H} = \mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}$ is called the **hat matrix**.

- The residual r_i is calculated by $\boxed{r_i = y_i - \hat{y}_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)}$ for $i = 1, \ldots, n$, or

$$\begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_n \end{bmatrix} \Rightarrow \boxed{\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y}}$$

Residuals | Mean & Variance

- In theory, the mean and variance of the residuals $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$ are:

$$\boxed{\mathrm{E}(\mathbf{r}) = \mathbf{0}} \quad \text{and} \quad \boxed{\mathrm{Var}(\mathbf{r}) = \sigma^2(\mathbf{I} - \mathbf{H})}$$

- Note: The model assumes $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$. Therefore, $\boxed{\mathrm{Var}(\mathbf{r}) \neq \mathrm{Var}(\boldsymbol{\varepsilon})}$.
  1. Denote the elements of $\mathbf{H}$ as:

$$\underset{n\times n}{\mathbf{H}} = \begin{bmatrix} h_{11} & \cdots & h_{1i} & \cdots & h_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ h_{i1} & \cdots & h_{ii} & \cdots & h_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ h_{n1} & \cdots & h_{ni} & \cdots & h_{nn} \end{bmatrix}$$

  2. $\mathrm{Var}(\epsilon_i) = \sigma^2$. However, $\mathrm{Var}(r_i) = \sigma^2(1 - h_{ii}) \Rightarrow$ The residuals have unequal variances.
  3. $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$. However, $\mathrm{Cov}(r_i, r_j) = -h_{ij}\sigma^2 \Rightarrow$ The residuals are negatively correlated.

- Numerical properties
  1. $0 \leq h_{ii} \leq 1$
  2. $\sum_{i=1}^{n} h_{ii} = \dim(\boldsymbol{\beta}) = p^*$, the number of model coefficients in $\boldsymbol{\beta}$

Standardized residuals

- The **standardized residuals** r_i^*'s are

$$r_i^* = \frac{r_i - \mathrm{E}(r_i)}{\mathrm{SE}(r_i)} = \frac{r_i - \mathrm{E}(r_i)}{\sqrt{\hat{\mathrm{Var}}(r_i)}} = \frac{r_i - 0}{\sqrt{\hat{\sigma}^2(1 - h_{ii})}} = \frac{r_i}{\hat{\sigma}\sqrt{1 - h_{ii}}} \quad \text{for } i = 1, \ldots, n$$

1. The standardized residuals r_i^*'s are approximately $\mathrm{N}(0, 1)$.
2. Therefore, 95% of r_i^*'s are within ± 1.96.
3. **Interpretation**: If $|r_i^*|$ is very large, the observation associated with this r_i^* can be considered an outlier.
4. In R, we use the `rstandard()` command to compute the standardized residuals r_i^*'s.

```
> m_fit <- lm(y ~ x, data = df_mydata)
> r     <- residuals(m_fit)          # raw residuals
> r_std <- rstandard(m_fit)          # standardized residuals
```

Leverage

- The **leverages** are the diagonal elements $h_{11}, \ldots, h_{ii}, \ldots, h_{nn}$ in the hat matrix $\mathbf{H}$.
  1. We have $\hat{y}_i = h_{i1}y_1 + \cdots + h_{ii}y_i + \cdots + h_{in}y_n$, because

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y} = \begin{bmatrix} h_{11} & \cdots & h_{1i} & \cdots & h_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ h_{i1} & \cdots & h_{ii} & \cdots & h_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ h_{n1} & \cdots & h_{ni} & \cdots & h_{nn} \end{bmatrix} \mathbf{y} \Rightarrow \hat{y}_i = \begin{bmatrix} h_{i1} & \cdots & h_{ii} & \cdots & h_{in} \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_n \end{bmatrix} = \sum_{j=1}^{n} h_{ij}y_j$$

  2. h_{ii} represents the amount of leverage a data point y_i has on its own predicted value $\hat{y}_i$.
  3. In R, we use the `hatvalues()` command to compute the leverages $h_{11}, \ldots, h_{ii}, \ldots, h_{nn}$.

```
> m_fit <- lm(y ~ x, data = df_mydata)
> hatvalues(m_fit)
```

Cook's Distance

- Calculate the predicted response $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ with all data.
- Calculate $\hat{\mathbf{y}}_{(-i)} = \mathbf{X}\hat{\boldsymbol{\beta}}_{(-i)}$, the predicted response using the model coefficient estimates that are calculated with all data **except** the observations from ith object.
  1. For SLR,

$$\underset{2\times 1}{\hat{\boldsymbol{\beta}}_{(-i)}} = \begin{bmatrix} \hat{\beta}_{(-i)0} \\ \hat{\beta}_{(-i)1} \end{bmatrix} = (\mathbf{X}_{[-i,\,]}^{\mathsf{T}}\mathbf{X}_{[-i,\,]})^{-1}\mathbf{X}_{[-i,\,]}^{\mathsf{T}}\mathbf{y}_{[-i]}, \quad \text{where } \underset{(n-1)\times 2}{\mathbf{X}_{[-i,\,]}} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_{i-1} \\ 1 & x_{i+1} \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \text{ and } \underset{(n-1)\times 1}{\mathbf{y}_{[-i]}} = \begin{bmatrix} y_1 \\ \vdots \\ y_{i-1} \\ y_{i+1} \\ \vdots \\ y_n \end{bmatrix}$$

  2. Note: $\hat{\mathbf{y}}$ is an $n \times 1$ vector of the predicted responses using $\hat{\boldsymbol{\beta}}$, and $\hat{\mathbf{y}}_{(-i)}$ is an $n \times 1$ vector of the predicted responses using $\hat{\boldsymbol{\beta}}_{(-i)}$.

Cook's Distance

- **Cook's Distance** is a standardized distance between $\hat{\mathbf{y}}$ and $\hat{\mathbf{y}}_{(-i)}$: For $i = 1, \ldots, n$,

$$D_i = \frac{\sum_{j=1}^{n} (\hat{y}_j - \hat{y}_{(-i)j})^2}{p^* \hat{\sigma}^2} = \frac{(\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(-i)})^\intercal (\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(-i)})}{p^* \hat{\sigma}^2} \quad \text{with } p^* = \dim(\boldsymbol{\beta})$$

1. **Interpretation**: Suppose ith object is left out. If the predicted responses have a large amount of changes (i.e., $\sum_{j=1}^{n} (\hat{y}_j - \hat{y}_{(-i)j})^2$ is large), then D_i becomes large. The data point with a large value of D_i is an influential point.
2. In practice, we do not repeat n different predictions $\hat{\mathbf{y}}_{(-1)}, \ldots, \hat{\mathbf{y}}_{(-n)}$, because

$$D_i = \frac{\sum_{j=1}^{n} (\hat{y}_j - \hat{y}_{(-i)j})^2}{p^* \hat{\sigma}^2} = \frac{r_i^2 h_{ii}}{p^* \hat{\sigma}^2 (1 - h_{ii})^2} = \frac{\left(\frac{r_i}{\hat{\sigma}\sqrt{1 - h_{ii}}} \right)^2 h_{ii}}{p^* (1 - h_{ii})} \Rightarrow \boxed{D_i = \frac{r_i^{*2} h_{ii}}{p^* (1 - h_{ii})}}$$

3. Note: If $h_{ii} \to 0$, then $D_i \to 0$. If $h_{ii} \to 1$, then $D_i \to \infty$.
4. In R, we use the `cooks.distance()` command to compute Cook's Distances $D_1, \ldots, D_n$.

```
> m_fit <- lm(y ~ x, data = df_mydata)
> cooks.distance(m_fit)
```

Three assumptions on the model errors

- We assumed $\epsilon_i \overset{\text{iid}}{\sim} \text{N}(0, \sigma^2)$ for SLR and $\epsilon_{ij} \overset{\text{iid}}{\sim} \text{N}(0, \sigma^2)$ for one-way ANOVA.
- **Independence**: Check the independence of the standardized residuals r_i^*'s for time series using the `acf()` command.
- **Equal variance** (homoscedasticity):
  1. Plot of $(\hat{y}_i, r_i^*)$'s should not show any pattern.
  2. Scale-location plot is a plot of $(\hat{y}_i, \sqrt{|r_i^*|})$'s: The smoothing red line should be approximately horizontal. Also, the spread around the smoothing red line should not vary much with the predicted responses.
  3. To test for heteroscedasticity in a regression model, we can do a Breusch-Pagan Test.

```
> library(lmtest)
> m_reg <- lm(y ~ x, data = df_mydata)
> bptest(m_reg)        # H_0: Residual variance does not depend on predictor x
```

  4. To test for heteroscedasticity in an ANOVA model, we can use a Levene's Test.

```
> library(car)
> m_anova <- lm(phos ~ variety, data = df_phos)
> leveneTest(m_anova)  # H_0: Equal variances for different varieties
```

Three assumptions on the model errors

- **Normality**: We check the normality of the model errors based on the standardized residuals r_i^*'s.
  1. Histogram of standardized residuals should show a standard normal distribution.

```
> r_std <- rstandard(m_fit)          # standardized residuals
> hist(r_std)                        # histogram of standardized residuals
```

  2. Q-Q plot of standardized residuals should show a good match between sample quantiles and their corresponding theoretical quantiles, displaying a strong linear relation.

```
> qqnorm(r_std)                      # Q-Q plot for normality check
> qqline(r_std)                      # add a reference line
```

  3. Normality tests, including Shapiro-Wilk test and Kolmogorov-Smirnov test, should **not** reject H_0: the standardized residuals are normally distributed.

```
> shapiro.test(r_std)                # Shapiro-Wilk normality test
> ks.test(r_std, "pnorm")            # Kolmogorov-Smirnov normality test
```